

Modeling Climate Anomalies
with Statistical Analysis

Today's goals



Module 1

Introduction to Python for Data Analysis

Course Introduction

- ✓ Course Updates and Accessibility Support
Reading • 1 min
- ✓ Earn Academic Credit for your Work!
Reading • 10 min
- ✓ Course Support
Reading • 10 min
- Introduction to the Course
Video • 2 min
- Meet the Instructor
Video • 1 min
- ✓ Introduction to Pandas for Data Loading and
Exploration
Video • 25 min
- ✓ Pandas
Reading • 15 min
- ✓ Course Assignments with Jupyter
Reading • 10 min

Visualizing Data with Python

- ✓ Introduction to Matplotlib and Seaborn for Data
Visualization
Video • 25 min

Module Assignment

- ✓ Introduction to Pandas and Matplotlib
Programming Assignment • Grade: 100%
- ✓ Pandas and Matplotlib
Graded Assignment • Grade: 93.33%

Module 2

Collecting Climate Data

Data Collection and APIs

- ✓ Introduction to APIs
Video • 9 min
- ✓ APIs, JSON, and Python Requests
Reading • 15 min
- ✓ API Hands-On (Lab Activity)
Reading • 30 min
- ✓ API Hands-On
Graded Assignment • Grade: 87.5%

Data Sources: NOAA

- ✓ Introduction to NOAA Website for Data
Collection
Video • 9 min
- ✓ NOAA API and Available Data
Reading • 10 min

Data Sources: USGS

Introduction to USGS Website and Available Data

Methods and Datasets via NWIS



0:09 / 0:38



2x



Dataretrieval

dataretrieval is a python alternative to USGS-R's dataRetrieval package for obtaining USGS or EPA water quality data, streamflow data, and meta data directly from web services. Note that dataretrieval is an **alternative** to theR package, not a port, in that it reproduces and expands upon the functionality of the R package but the organization and functionality often differs.

[dataretrieval-python](#)  (external link; opens in a new tab)

USGC Parameter Codes

To find the available data offered from USGS, refer to the following code definitions.

[Parameter Code Definition](#)  (external link; opens in a new tab)

Go to next item

✓ Completed



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